MATH 39 — Linear Algebra

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Contents

1. Systems of Linear Equations and Matrices

1.1 Linear Equations

Definition A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \ldots, a_n$ and b are constants. The constants $a_1, a_2, \ldots, a_n$ are called the **coefficients** of the equation, and b is called the **constant term**.

Examples of linear equations include:

- $2x + 3y - z = 5$
- $-x_1 + 4x_2 + 0x_3 = 7$
- $0x + 0y + 0z = 0$

Non-examples of linear equations include:

- $x^2 + y = 3$ (because of the x^2 term)
- $xy + z = 4$ (because of the xy term)
- $\sin(x) + y = 1$ (because of the $\sin(x)$ term)
- $x + \sqrt{y} = 2$ (because of the $\sqrt{y}$ term)

Systems of Linear Equations

Definition A **finite** collection of linear equations involving the same set of variables is called a **system of linear equations**.

Example of a system of linear equations:

$$\begin{cases} 2x + 3y - z = 5 \\ -x + 4y + 2z = 7 \\ 3x - y + z = 2 \\ \cdots \end{cases}$$

Definition A sequence of numbers $(s_1, s_2, \ldots, s_n)$ is called a **solution** to the system of linear equations if, when we substitute s_i for x_i in each equation, the equation holds true.

Example Consider the system of equations:

$$\begin{cases} x + y + z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

Is the sequence $(-2, -5, 0)$ a solution to this system? Does $x = y = z = 1$?

Solution To check if $(-2, -5, 0)$ is a solution, substitute $x = -2$, $y = -5$, and $z = 0$ into both equations:

- First equation: $(-2) + (-5) + 0 = -7 \neq 3$ (does not hold)
- Second equation: $2(-2) + (-5) + 3(0) = -4 - 5 = -9 \neq 1$ (does not hold)

Thus, $(-2, -5, 0)$ is not a solution to the system.
Next, check if $(1, 1, 1)$ is a solution:

- First equation: $1 + 1 + 1 = 3$ (holds)
- Second equation: $2(1) + 1 + 3(1) = 2 + 1 + 3 = 6 \neq 1$ (does not hold)

Thus, $(1, 1, 1)$ is also not a solution to the system.

Solutions to Linear Systems

Definition There are three possible scenarios for the solutions to a system of linear equations:

- **No solution**: The system is inconsistent, meaning there are no values for the variables that satisfy all equations simultaneously.
- **Unique solution**: There is exactly one set of values for the variables that satisfies all equations.
- **Infinitely many solutions**: There are multiple sets of values for the variables that satisfy all equations. (i.e. the same line or scalar multiples of each other)

Systems of linear equations that have at least one solution are called **consistent**, while those with no solutions are called **inconsistent**.

Matricies

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. The numbers in the matrix are called its **entries** or **elements**. A matrix with m rows and n columns is called an $m \times n$ matrix.

Example of an **augmented** matrix:

$$\begin{cases} 3x_1 + 2x_2 - x_3 + x_4 = -1 \\ 2x_1 - x_3 + 2x_4 = 0 \\ 3x_1 + x_2 + 2x_3 + 5x_4 = 2 \end{cases} = \begin{bmatrix} 3 & 2 & -1 & 1 & | & -1 \\ 2 & 0 & -1 & 2 & | & 0 \\ 3 & 1 & 2 & 5 & | & 2 \end{bmatrix}$$

Elementary Operations

Definition There are three types of **elementary row operations** that can be performed on a matrix:

- **Row swapping**: Interchanging two rows of the matrix.
- **Row scaling**: Multiplying all entries in a row by a non-zero constant.
- **Row addition**: Adding a multiple of one row to another row.

These operations are used in methods such as Gaussian elimination to solve systems of linear equations. Row operations do not produce equal matricies, but they do produce **equivalent** matrices, meaning they represent systems of equations with the same solution set. In order for 2 matricies to be equal, they must have the same dimensions and identical entries in corresponding positions.

Example Consider the matricies:

$$\begin{bmatrix} 1 & 2 & | & 0 \\ 3 & 4 & | & 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 & 4 & | & 0 \\ 3 & 4 & | & 1 \end{bmatrix}$$

This represents the row operation of scaling the first row by 2 ($2R_1 \mapsto R_1$). These matricies are equivalent but not equal.

Example Solve:

$$\begin{cases} 2x + 3y - z = 3 \\ -x - y + z = 1 \\ x - 2y - 5z = 0 \end{cases} = \begin{bmatrix} 2 & 3 & -1 & | & 3 \\ -1 & -1 & 1 & | & 1 \\ 1 & -2 & -5 & | & 0 \end{bmatrix} \xmapsto{R3 \mapsto R1} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ -1 & -1 & 1 & | & 1 \\ 2 & 3 & -1 & | & 3 \end{bmatrix} \xmapsto{R2+R1 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & -3 & -4 & | & 1 \\ 2 & 3 & -1 & | & 3 \end{bmatrix}$$

$$\xmapsto{R3-2R1 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & -3 & -4 & | & 1 \\ 0 & 7 & 9 & | & 3 \end{bmatrix} \xmapsto{-\frac{1}{3}R2 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 7 & 9 & | & 3 \end{bmatrix} \xmapsto{R3-7R2 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 0 & \frac{-1}{3} & | & \frac{16}{3} \end{bmatrix}$$

$$\xmapsto{-3R3 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \xmapsto{R2-\frac{4}{3}R3 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \xmapsto{R1+5R3 \mapsto R1} \begin{bmatrix} 1 & -2 & 0 & | & -80 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix}$$

$$\xmapsto{R1+2R2 \mapsto R1} \begin{bmatrix} 1 & 0 & 0 & | & -38 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \implies (x, y, z) = (-38, 21, -16)$$

1.2 Gaussian Elimination

Gaussian elimination is a method for solving systems of linear equations by transforming the system's augmented matrix into a simpler form using elementary row operations. The goal is to reach **row echelon form** or **reduced row echelon form**, from which the solutions can be easily obtained.

Row Echelon Form

Definition A matrix is in **row echelon form** if it satisfies the following conditions:

- All non-zero rows are above any rows of all zeros.
- The leading entry (the first non-zero number from the left, also called a **pivot**) of each non-zero row is to the right of the leading entry of the previous row.
- The leading entry in any non-zero row is 1 (this condition is sometimes omitted in basic row echelon form).

Example of a matrix in row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

It is often useful to further refine this to **reduced row echelon form**, which additionally requires that the leading 1 in each non-zero row is the only non-zero entry in its **column**.

Example of a matrix in reduced row echelon form:

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Notes

- Every matrix can be reduced to row echelon form and reduced row echelon form using a sequence of elementary row operations.
- The reduced row echelon form of any matrix is unique.
- Row echelon form is not unique; different sequences of row operations can lead to different row echelon forms for the same matrix.

Zero Matrix A **zero matrix** is a matrix in which all entries are zero. It is often denoted by 0. The solution to the zero matrix is $\mathbb{R}^n$, where n is the number of variables.

Matrix Reduction Algorithm

1. Identify the leftmost non-zero column (the pivot column).
2. Select a non-zero entry in the pivot column as the pivot. If necessary, swap rows to move this entry to the top of the pivot column.

3. Scale the pivot row so that the pivot entry becomes 1.

4. Use row addition to create zeros in all positions below the pivot.

5. Cover the row containing the pivot and repeat the process for the submatrix that remains until all rows have been processed.

6. (Optional) To achieve reduced row echelon form, repeat the process to create zeros above each pivot as well.

Gaussian Elimination Algorithm

1. Augemnted Matrix -¿ RREF with Row Operations

2. If a row of the form [0 0 ... 0 — b] with b non-zero exists, then the system is inconsistent (no solutions).

3. Otherwise, assign free variables (if any) and solve for leading variables to find the solution set.

4. Write leading variables in terms of the parameters

5. If there are no nonleading variables, we gave 1 solution. If there are nonleading variables, we have infinitely many solutions.

Example 1 Solve the system:

$$\begin{cases} 3x + y - 4z = 1 \\ x + 10z = 5 \\ 4x + y + 6z = 1 \end{cases} = \left[\begin{array}{ccc|c} 3 & 1 & -4 & 1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right] \xmapsto{R2 \leftrightarrow R1} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & 1 \\ 4 & 1 & 6 & 1 \end{array}\right] \xmapsto{R2-3R1 \mapsto R2} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

$$\xmapsto{R3-4R1 \mapsto R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 1 & -34 & -19 \end{array}\right] \xmapsto{R3-R2 \mapsto R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 0 & 0 & -5 \end{array}\right] \implies \text{no solutions}$$

Example 2 Solve the system:

$$\begin{cases} x_1 - 2x_2 - x_3 + 3x + 4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases} = \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xmapsto{R2-2R1 \mapsto R2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right]$$

$$\xmapsto{R3-R1 \mapsto R3} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right] \xmapsto{R3-R2 \mapsto R3} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xmapsto{\frac{1}{3}R2 \mapsto R2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ x_3 - 2x_4 = 1 \end{cases}$$

$$\implies \begin{cases} x_2 = s \\ x_4 = t \\ x_1 = 1 + 2s + t \\ x_3 = 1 + 2t \end{cases}$$

Rank of a Matrix

Definition The **rank** of a matrix is the number of leading 1's (pivots) in its row echelon form or reduced row echelon form. It represents the dimension of the column space (or row space) of the matrix, which is the span of its columns (or rows). Rank is useful for finding the number of parameters in the solution set of a linear system.

Example 3 Consider the matrix from Example 2. Find the rank and number of parameters:

$$A = \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xmapsto{\text{RREF(A)}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \implies \text{rank}(A) = 2$$

$$n = 4 \text{ (number of variables)} \implies \text{number of parameters } = n - \text{rank}(A) = 4 - 2 = 2$$

Example 4 Find the rank of the following matricies:

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \implies \text{rank}(A) = 2$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} \xmapsto{R1 \leftrightarrow R2} \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \xmapsto{\frac{1}{3}R1 \mapsto R1} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \implies \text{rank}(B) = 1$$

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \text{rank}(C) = 0$$

1.3 Homogeneous Systems

Definition A system is called **homogeneous** if all the constant terms are zero. In other words, a system of equations is homogeneous if it can be written in the form

$$\begin{cases} x - 5y = 0 \\ 2x + 3y + z = 0 \end{cases}$$

The soluttion set of a homogeneous system always contains the **trivial solution**, which is the solution where all variables are equal to zero. For example, in the system above, the trivial solution is $x = 0, y = 0, z = 0$. Any other solution is called a **nontrivial solution**.

Example Find any non-trivial solution

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 2 & 2 & 0 & 1 & 0 \\ 3 & 1 & 2 & -1 & 0 \end{array}\right] \xmapsto{R_2 \mapsto R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 3 & 1 & 2 & -1 & 0 \end{array}\right]$$

$$\xmapsto{R_3 \mapsto R_3 - 3R_1} \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 0 & 4 & -4 & 2 & 0 \end{array}\right] \xmapsto{R_3 \mapsto R_3 - R_2} \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right]$$

$$\xmapsto{R_1 \mapsto R_1 - R_3} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right] \xmapsto{R_2 \mapsto R_2 + 3R_3} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 4 & -4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right]$$

$$\xmapsto{R_3 \mapsto -R_3} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 4 & -4 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \xmapsto{R_2 \mapsto \frac{1}{4}R_2} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xmapsto{R_1 \mapsto R_1 + R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \implies \begin{cases} x_1 + x_3 = 0 \\ x_2 - x_3 = 0 \\ x_4 = 0 \end{cases} = \begin{cases} x_1 = -x_3 \\ x_2 = x_3 \\ x_4 = 0 \end{cases}$$

$$x_3 = S \implies \begin{cases} x_1 = -S \\ x_2 = S \\ x_3 = S \\ x_4 = 0 \end{cases} \quad S \in \mathbb{R} \qquad S = 1 \implies \begin{cases} x_1 = -1 \\ x_2 = 1 \\ x_3 = 1 \\ x_4 = 0 \end{cases}$$

Theorem If a homogeneous system of linear equations has ore variables than equations, then it has infinitly many non-trivial solutions. For some matrix A with m rows and n parameters $\text{rank}(A) \leq m < n$ implies that the solutions involved $n - rank(A) \geq 1$ parameters, so $x_i = S$ and we can choose any value for S other than zero to get a non-trivial solution.

Note If a homogeneous system has a nontrivial solution, it need not have more variables than equations.

Note If a homogeneous system has a non-trivial solution, then it has infinitely many non-trivial solutions. If $x = S$ is a non-trivial solution, then so is $x = kS$ for any scalar $k \neq 0$.

Vectors

Definition A **vector** in $\mathbb{R}^n$ is an ordered n-tuple of real numbers. We can represent a vector as a column matrix:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = (x_1, x_2, \ldots, x_n) = \langle x_1, x_2, \ldots, x_n \rangle$$

where $x_1, x_2, \ldots, x_n$ are the components of the vector $\vec{x}$. In this class, vectors should always be represented as column matrices unless otherwise specified.

Definition A vector of length n is an ordered n-tuple, and the set of all vectors of length n is denoted by $\mathbb{R}^n$. In other words, $\vec{x} \in \mathbb{R}^n$ if $\vec{x}$ has n components.

Vector Addition Vector addition can only be performed on vectors of the same length. If $\vec{x}, \vec{y} \in \mathbb{R}^n$, then their sum is given by

$$\vec{x} + \vec{y} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Scalar Multiplication If $\vec{x} \in \mathbb{R}^n$ and c is a scalar, then the scalar multiplication of c and $\vec{x}$ is given by

$$c\vec{x} = c \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$$

Linear Combination A sum of scalar multiples of several vectors or columns is called a **linear combination** of the vectors or columns. For example, if $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and a, b, c are scalars, then

$$a\vec{u} + b\vec{v} + c\vec{w}$$

is a linear combination of the vectors $\vec{u}, \vec{v}, \vec{w}$.

Example Let $\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{y} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \vec{z} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \vec{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.
a. Find $2\vec{x} - \vec{y} + 3\vec{z}$, b. Determine if $\vec{v}$ is a linear combination of $\vec{x}, \vec{y}, \vec{z}$, and c. Determine if $\vec{w}$ is a linear combination of $\vec{x}, \vec{y}, \vec{z}$.

$$a. \quad 2\vec{x} - \vec{y} + 3\vec{z} = 2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 3 \end{bmatrix}$$

$$b. \quad \exists r, s, t \in \mathbb{R} \ni r\vec{x} + s\vec{y} + t\vec{z} = \vec{v} \implies \begin{bmatrix} r \\ 0 \\ r \end{bmatrix} + \begin{bmatrix} 2s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} 3t \\ t \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

$$\implies \begin{cases} r + 2s + 3t = 0 \\ s + t = -1 \\ r + t = 2 \end{cases} \implies \begin{bmatrix} 1 & 2 & 3 & | & 0 \\ 0 & 1 & 1 & | & -1 \\ 1 & 0 & 1 & | & 2 \end{bmatrix} \xmapsto{RREF} \begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & 1 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\implies \begin{cases} r + t = 2 \\ s + t = -1 \end{cases} \implies \begin{cases} r = 2 - q \\ s = -1 - q \\ t = q \end{cases} \quad t \in \mathbb{R} \quad q = 0 \implies \begin{cases} r = 2 \\ s = -1 \\ t = 0 \end{cases} \implies \vec{v} = 2\vec{x} - \vec{y}$$

$$c. \quad \exists r, s, t \in \mathbb{R} \ni r\vec{x} + s\vec{y} + t\vec{z} = \vec{w} \implies \begin{bmatrix} 1,2,3 & | & 1 \\ 0,1,1 & | & 1 \\ 1,0,1 & | & 1 \end{bmatrix} \xmapsto{RREF} \begin{bmatrix} 1 & 0 & 1 & | & 1 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & -1 \end{bmatrix}$$

$\implies$ No solution, so $\vec{w}$ is not a linear combination of $\vec{x}, \vec{y}, \vec{z}$

Note To find if $\vec{x}$ is a linear combinition of $\vec{a_1}, \vec{a_2}, \ldots, \vec{a_n}$, we can form the augmented matrix

$$[\vec{a_1} \ \vec{a_2} \ \ldots \ \vec{a_n} \mid \vec{x}]$$

and row reduce to see if there is a solution. Linear combinations are a good way to write the general solution to a homogeneous system.

Theorem Any linear combination of solutions to a homogeneous system is also a solution to the system.

Example Let $\vec{p} = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$ be solutions to the homogeneous system $a_1x_1 + a_2x_2 = 0$, hence:

$$\implies \exists s, t \in \mathbb{R} \ni \begin{cases} a_1p_1 + a_2p_2 = 0 \\ a_1q_1 + a_2q_2 = 0 \end{cases} = s\vec{p} + t\vec{q} = \begin{bmatrix} sp_1 + tq_1 \\ sp_2 + tq_2 \end{bmatrix} \text{ is a solution}$$

$$a_1(sp_1 + tq_1) + a_2(sp_2 + tq_2) = 0 \implies s \cdot 0 + t \cdot 0 = 0$$

Example Solve the homogeneous system w/ coefficient matrix $A = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -3 & 6 & 1 & 0 \\ -2 & 4 & 4 & -2 \end{bmatrix}$

$$RREF(A) = \begin{bmatrix} 1 & -2 & 0 & -\frac{1}{5} & | & 0 \\ 0 & 0 & 1 & -\frac{3}{5} & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \quad \text{Rank}(A) = 2 \implies \text{4 - 2} = 2 \text{ parameters}$$

$$\implies \begin{cases} x_1 = 2s + \frac{1}{5}t \\ x_2 = s \\ x_3 = \frac{3}{5}t \\ x_4 = t \end{cases} = \begin{bmatrix} 2s + \frac{1}{5}t \\ s \\ \frac{3}{5}t \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix} \text{ is the general solution,}$$

so our basic solutions are $\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$

Note Every solution to a homogeneous system can be written as a linear combination of the basic solutions.

2. Matricies and Matrix Operations

2.1 Matricies and Matrix Operations

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns. For example, the following is a 2×3 matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Matrix Access The entry in the i-th row and j-th column of a matrix A is denoted by a_{ij}. For example, in the matrix A above, $a_{12} = 2$ and $a_{21} = 4$.

$$A_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Matrix Operations

Matrix Equality Two matrices A and B are equal if they have the same dimensions and their corresponding entries are equal. That is, $A = B$ if and only if $a_{ij} = b_{ij}$ for all i and j.

Matrix Addition Two matrices can be added if they have the same dimensions, otherwise it is undefined. If A and B are both $m \times n$ matrices, then their sum is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

Scalar Multiplication If A is an $m \times n$ matrix and c is a scalar, then the scalar multiplication of c and A is given by

$$cA = \begin{bmatrix} ca_{11} & ca_{12} & \dots & ca_{1n} \\ ca_{21} & ca_{22} & \dots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \dots & ca_{mn} \end{bmatrix}$$

Properties of Matrix Addition and Scalar Multiplication Let A, B, and C be matrices of the same dimensions, and let c and d be scalars. Then the following properties hold:

- Commutative Property: $A + B = B + A$
- Associative Property: $(A + B) + C = A + (B + C)$
- Distributive Property: $c(A + B) = cA + cB$
- Scalar Distributive Property: $(c + d)A = cA + dA$

- Associative Property of Scalar Multiplication: $c(dA) = (cd)A$
- Identity Property: $A + 0 = A$, where 0 is the zero matrix of the same dimensions as A

Matrix Transpose The transpose of a matrix A, denoted by A^T, is obtained by interchanging its rows and columns. If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix given by

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}$$

Transposition Identities Let A, B be $m \times n$ matricies and $k \in \mathbb{R}$

- A^T is $n \times m$
- $(A^T)^T = A$
- $(kA)^T = k(A^T)$
- $(A + B)^T = A^T + B^T$

Example Solve for A

$$(2A^T + 3 \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix})^T$$

Symmetric Matrix A matrix A is called symmetric if $A = A^T$. If A, B are $n \times n$ symmetric matrices, then $A + B$ is also symmetric. If c is a scalar, then cA is also symmetric. If A and B are both symmetric, then $kA + kB$ is also symmetric for $k, p \in \mathbb{R}$.

2.2 Matrix-Vector Multiplication

Vector-Matrix Multiplication

Definition Let A be an $m \times n$ matrix written $A = [a_1\ a_2\ \dots\ a_n]$ where a_i is the i-th column of A. Let $\vec{x}$ be an $n \times 1$ column vector written as $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$. Then the matrix-vector product $A\vec{x}$ is defined as

$$\begin{aligned} A\vec{x} &= [a_1\ a_2\ \dots\ a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= x_1a_1 + x_2a_2 + \ldots + x_na_n \\ &= \sum_{i=1}^{n} x_ia_i \end{aligned}$$

Notes

- If A is an $m \times n$ matrix, then $A\vec{x}$ is defined only for $\vec{x} \in \mathbb{R}^n$.
- The inner sizes of the vector and matrix must match for the product to be defined.
- $A\vec{x}$ is a linear combination of the columns of A with coefficients x_i in order.

Example Let $B = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Find $B\vec{q}$.

$$B\vec{q} = 1 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \end{bmatrix}$$

Example Let $A = \begin{bmatrix} 0 & -1 & 2 \\ 5 & 3 & -4 \\ 1 & 0 & 1 \end{bmatrix}$ and $\vec{p} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$. Find $A\vec{p}$.

$$A\vec{p} = 1 \cdot \begin{bmatrix} 0 \\ 5 \\ 1 \end{bmatrix} + 0 \cdot \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + (-3) \cdot \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -6 \\ 12 \\ -3 \end{bmatrix} = \begin{bmatrix} -6 \\ 17 \\ -2 \end{bmatrix}$$

Theorem A system of linear equations $\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m \end{cases}$ can be written in matrix form as $A\vec{x} = \vec{b}$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

A is called the coefficient matrix, $\vec{x}$ is called the variable vector, and $\vec{b}$ is called the constant vector. The system of equations has a solution if and only if there exists a vector $\vec{x}$ such that $A\vec{x} = \vec{b}$.

- Every system of linear equations can be written as $A\vec{x} = \vec{b}$
- $A\vec{x} = \vec{b}$ is consistant if and only if $\vec{b}$ is a linear combination of columns of A.

Identitiy Matrix

Definition The $n \times n$ identity matrix, denoted by I_n, is the square matrix with 1's on the main diagonal and 0's elsewhere. That is,

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

Properties of the Identity Matrix

- $\text{rank}(I_n) = n$
- $I_n\vec{x} = \vec{x}$ for all $\vec{x} \in \mathbb{R}^n$

More Indentities Let A and B be $n \times n$ matricies and $\vec{x}, \vec{y} \in \mathbb{R}^n$

- $A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y}$
- $A(k\vec{x}) = k(A\vec{x})$ for $k \in \mathbb{R}$
- $(A + B)\vec{x} = A\vec{x} + B\vec{x}$

Theorem Let $A\vec{x} = \vec{b}$ be a system of linear equations and $\text{rank}(A) = r$. Then the rank of the augmented matrix $[A|\vec{b}]$ is either r or $r + 1$. The system is consistent if and only if $\text{rank}(A) = \text{rank}([A|\vec{b}])$.

Example Let $A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. Determine if the system $A\vec{x} = \vec{b}$ is consistent.

$$\text{rank}(A) = 2 \text{ and } \text{rank}([A|\vec{b}]) = 3 \implies \text{the system is inconsistent}$$

Dot Product Let $\vec{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$ be two vectors in $\mathbb{R}^n$. The dot product of $\vec{a}$ and $\vec{b}$, denoted by $\vec{a} \cdot \vec{b}$, is defined as

$$\vec{a} \cdot \vec{b} = a_1b_1 + \ldots + a_nb_n = \sum_{i=1}^{n} a_ib_i$$

The result of the dot product is a scalar.

Definition Let A be an $m \times n$ matrix and $\vec{x} \in \mathbb{R}^n$, then the ith entry of the vector $A\vec{x}$ is the dot product of the ith row of A with $\vec{x}$

Example 2Let $A = \begin{bmatrix} 1 & -1 & 3 & -2 \\ 2 & 3 & 5 & 7 \\ 3 & 0 & 4 & 0 \\ 4 & 5 & 2 & 0 \end{bmatrix}$ and $\vec{x} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 5 \end{bmatrix}$. Find the second entry of $A\vec{x}$ without computing the entire product.

$$P_2 = [2\ 3\ 5\ 7] \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \\ 5 \end{bmatrix} = 2(0) + 3(1) + 5(-1) + 7(5) = 3 - 5 + 35 = 33$$

Example Let $C = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ -1 \end{bmatrix}$. Find $C\vec{q}$.

$$\begin{aligned} C\vec{q} &= 2 \cdot \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} + 1 \cdot \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + 0 \cdot \begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 0 \\ -6 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -5 \\ -1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix} \end{aligned}$$

Standard Basis Vectors The standard basis vectors in $\mathbb{R}^n$ are the vectors $e_1, e_2, \ldots, e_n$ where e_i has a 1 in the i-th position and 0's elsewhere. That is,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad e_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

We can use these basis vectors to isolate the columns of a matrix. For example, if $A = [a_1\ a_2\ \ldots\ a_n]$ is an $m \times n$ matrix, then

$$\begin{aligned} Ae_1 &= a_1 \\ Ae_2 &= a_2 \\ &\vdots \\ Ae_n &= a_n \end{aligned}$$

Theorem $A, B \in \mathbb{R}^{m \times n}, \vec{x} \in \mathbb{R}^n \implies A = B \implies \forall i < n \in \mathbb{N}, Ae_i = Be_i$

Note $A\vec{x} = B\vec{x}$ does not imply $A = B$ for all $\vec{x} \in \mathbb{R}^n$. For example, if A and B are both $m \times n$ matrices with all entries equal to 0, then $A\vec{x} = B\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$, but A and B are not necessarily the same matrix.

Theorem $A\vec{x} = B\vec{x} \implies \forall \vec{x} \in \mathbb{R}^n, A\vec{x} = B\vec{x}$

Transformations

Definition Functions $T : \mathbb{R}^n \to \mathbb{R}^m$ are called transformations. The set $\mathbb{R}^n$ is called the domain of T and the set $\mathbb{R}^m$ is called the codomain of T. The image of a vector $\vec{x} \in \mathbb{R}^n$ under the transformation T is denoted by $T(\vec{x})$.

Matrix Transformations A matrix transformation is denoted by $T_A(\vec{x}) = A\vec{x}$ for some matrix $A_{m\times n}$. If $A \in \mathbb{R}^{m\times n}$, then $T_A : \mathbb{R}^n \to \mathbb{R}^m$

Example $\forall \vec{x} \in \mathbb{R}^2, T_A(\vec{x}) = A\vec{x}$ and let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Find $T_A(\vec{u})$.

$$T_A(\vec{u}) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(2) + (-3)(-1) \\ 3(2) + 5(-1) \\ -1(2) + 7(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

Example Let $\vec{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$. Find $\vec{x} \in \mathbb{R}^2 \ni T_A(\vec{x}) = \vec{b}$ and $\vec{y} \in \mathbb{R}^2 \ni T_A(\vec{y}) = \vec{c}$.

$$A\vec{x} = \vec{b} \implies \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} \implies \begin{bmatrix} 1 & -3 & | & 3 \\ 3 & 5 & | & 2 \\ -1 & 7 & | & -5 \end{bmatrix} \xmapsto{RREF} \begin{bmatrix} 1 & 0 & | & \frac{3}{2} \\ 0 & 1 & | & -\frac{1}{2} \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\implies \vec{x} = \begin{bmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$A\vec{y} = \vec{c} \implies \begin{bmatrix} 1 & -3|3 \\ 3 & 5|2 \\ -1 & 7|5 \end{bmatrix} \xmapsto{RREF} \begin{bmatrix} 1 & -3 & | & 3 \\ 0 & 1 & | & -\frac{1}{2} \\ 0 & 0 & | & 10 \end{bmatrix} \implies \text{the system is inconsistent}$$

2.3 Matrix-Matrix Multiplication

Definition $\forall A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times p} \implies AB \in \mathbb{R}^{m\times p} = \begin{bmatrix} A\vec{b_1} & A\vec{b_2} & \dots & A\vec{b_p} \end{bmatrix}$ where $\vec{b_i}$ is the i-th column of B.

Properties

- $AB \neq BA$
- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- $(A + B)C = AC + BC$
- $k(AB) = (kA)B = A(kB)$ for any scalar k
- $(AB)^T = B^T A^T$

Block Multiplication

Definition We can partition A and B into blocks and perform multiplication on the blocks. For example, if we partition A into four blocks and B into four blocks, we can write:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Then, the product AB can be computed as:

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

This method is particularly useful for large matrices, as it allows us to break down the multiplication into smaller, more manageable pieces.

Theorem If $A, B \in \mathbb{R}^{n\times n}$ are partitioned into comparable blocks, then the product AB can be computed by matrix multiplication using blocks as entries. In other words, if A and B are partitioned into blocks of size $m \times m$, then the product AB can be computed by multiplying the corresponding blocks and summing the results.

Multiplying Diagonal Matricies If A and B are diagonal matrices, then their product is also a diagonal matrix. Specifically, if $A = \text{diag}(a_1, a_2, \dots, a_n)$ and $B = \text{diag}(b_1, b_2, \dots, b_n)$, then their product is given by:

$$AB = \begin{bmatrix} a_1b_1 & 0 & \dots & 0 \\ 0 & a_2b_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_nb_n \end{bmatrix}$$

2.4 Matrix Inverses

Multiplicative Inverse If A is a square matrix and $AB = BA = I$, then B is called the multiplicative inverse of A and is denoted by $A^{-1} = B$.

Note Let $A \in \mathbb{R}^{m\times n} \ni m \neq n$ and $B \in \mathbb{R}^{n\times m}$, it is possible for $A \cdot B = I_m$ but $B \cdot A \neq I_n$. In this case, A is called a left inverse of B and B is called a right inverse of A.

Properties Let $A\vec{x} = \vec{b}$, then

- $A^{-1}A\vec{x} = A^{-1}\vec{b} \implies \vec{x} = A^{-1}\vec{b}$
- $I\vec{x} = A^{-1}\vec{b}$
- $\vec{x} = A^{-1}\vec{b}$

Theorem If A is invertible, then $A\vec{x} = \vec{0}$ has 1 unique solution.

Example Show that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ is the inverse of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$BA = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Example Show the $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ has no inverse.

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+2c & b+2d \\ 0 & 0 \end{bmatrix} \neq I$$

Determinants of 2x2 Matricies

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det(A) = ad - bc$. If $\det(A) \neq 0$, then A is invertible and $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Theorem $A\vec{x} = \vec{b}$ and $\exists A^{-1}$, then $\vec{x} = A^{-1}\vec{b}$

Example

Matrix Inversion Algorithm

If $A_{n\times n}$, then there is a sequence of elementary row operations that carry $A \to I_n$. The same sequence of elementary row operations carries $I_n \to A^{-1}$.

1. Form the augmented matrix $[A|I_n]$.
2. Use elementary row operations to reduce the left half of the augmented matrix to I_n.
3. The right half of the augmented matrix is now A^{-1}.
4. If the left half of the augmented matrix cannot be reduced to I_n, then A is not invertible.

Example

Properties of Inverses

Let A be invertible, then

- $AB = AC \implies B = C$
- $(A^T)^{-1} = (A^{-1})^T($
- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $AB(AB)^{-1} = AB(B^{-1}A^{-1}) = I$
- $[A_1\ A_2\ \ldots\ A_k]^{-1} = [A_k^{-1}\ \ldots\ A_2^{-1}\ A_1^{-1}]$
- $(A^k)^{-1} = (A^{-1})^k \forall k \in \mathbb{R}, k > 0$
- $(aA)^{-1} = \frac{1}{a}A^{-1} \forall a \in \mathbb{R}, a \neq 0$

Example

Inverse Theorem The following conditions are equivilent for an $n \times n$ matrix A:

- A is invertible
- $RREF(A) = I_n$
- $\text{rank}(A) = n$
- The homogeneous system $A\vec{x} = \vec{0}$ has only the trivial solution $\vec{x} = \vec{0}$
- The system $A\vec{x} = \vec{b}$ has a unique solution for every $\vec{b} \in \mathbb{R}^n$
- $det(A) \neq 0$
- The linear transformation $T(\vec{x}) = A\vec{x}$ is one-to-one and onto.

Inverses of Block Matricies Let $P = \begin{bmatrix} A_{m\times m} & X \\ 0 & B_{n\times n} \end{bmatrix}$ and $Q = \begin{bmatrix} A_{m\times m} & 0 \\ Y & B_{n\times n} \end{bmatrix}$, then $P^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}XB^{-1} \\ 0 & B^{-1} \end{bmatrix}$ and $Q^{-1} = \begin{bmatrix} A^{-1} & 0 \\ -B^{-1}YA^{-1} & B^{-1} \end{bmatrix}$

Theorem A matrix transformation T_A is invertible if and only if the matrix A is invertable. T_A being invertible means T is onto and one to one.

$$T_A(\vec{x}) = A\vec{x} \implies T_A^{-1}(\vec{y}) = A^{-1}\vec{y}$$

2.5 Elementary Matrices

Definition An $n \times n$ matrix E is an elementary matrix if it can be obtained from I_n by performing a single elementary row operation.

Types of Elementary Matrices

- Type 1: E is obtained from I_n by interchanging two rows.
- Type 2: E is obtained from I_n by multiplying a row by a nonzero scalar.
- Type 3: E is obtained from I_n by adding a scalar multiple of one row to another row.

Example

Theorem Let E be an elementary matrix and A be any matrix, then EA is the matrix obtained from A by performing the same elementary row operation on A as was performed on I_n to obtain E. Note that this only works for left multiplication, AE does not have the same property. You also cannot perform multiple elementary row operations by multiplying by multiple elementary matrices, you can only perform one elementary row operation by multiplying by one elementary matrix.

Example

Theorem All elementary matricies are invertible and the inverse of an elementary matrix is also an elementary matrix.

Example Suppose $A, B \in \mathbb{R}^{m\times n}$ are matrices and $A \xmapsto{ERO} B$, then

1. $B = UA$, where $U \in \mathbb{R}^{m\times m}$ is an invertible matrix.
2. U can be computed by $[A|I_m] \xmapsto{ERO} [B|U]$.
3. $U = E_k E_{k-1} \ldots E_1$, where E_i is the elementary matrix corresponding to the i-th elementary row operation performed on A to obtain B.

Example

Theorem A square matrix is invertible if and only if it is a product of elementary matricies.

Theorem If $A = E_1 E_2 \ldots E_k$, then $\det(A) = \det(E_1)\det(E_2)\ldots\det(E_k)$, where E_i is an elementary matrix. This is because the determinant of a product of matrices is the product of their determinants. Since each E_i is an elementary matrix, we can compute its determinant based on the type of elementary row operation it represents.

3. Section 3

3.1 Skipped

3.2 Determinants

Theorem Let A be an $n \times n$ matrix

1. If A has a row or column of zeros, then $det(A) = 0$
2. If two distinct rows or columns are interchanged, the determinant changes sign
3. If a row is multiplied by a scalar k, then the determinant is multiplied by k
4. $det(kA) = k^n det(A)$ and $det(-A) = -det(A)$
5. If two rows or columns are identical or scalar multiples of each other, then $det(A) = 0$
6. If a multiple of one row or column is added to another row or column, the determinant doesn't change

Example Find the determinant of $\begin{vmatrix} 1 & 1 & 1 & 2 \\ 2 & 2 & 0 & 1 \\ 3 & 3 & -1 & -3 \\ 4 & 2 & -5 & 7 \end{vmatrix}$

$$\det = \begin{vmatrix} 1 & 1 & 1 & 2 \\ 2 & 2 & 0 & 1 \\ 3 & 3 & -1 & -3 \\ 4 & 2 & -5 & 7 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 0 & 1 \\ 0 & 3 & -1 & -3 \\ -2 & 2 & -5 & 7 \end{vmatrix} \qquad = -2 \begin{vmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 3 & -1 & -3 \end{vmatrix} = -2(-1) \begin{vmatrix} 2 & 1 \\ 4 & -1 \end{vmatrix} \qquad = 2(-2-4) = -12$$

Example $\det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} = 6$. Evaluate $\det \begin{bmatrix} a+x & b+y & c+z \\ 3x & 3y & 3z \\ -p & -q & -r \end{bmatrix}$

$$= -3(-1) \begin{vmatrix} a+x & b+y & c+z \\ x & y & z \\ p & q & r \end{vmatrix} = (-1)(-3) \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = 3 \cdot 6 = 18$$

Determinants of Elementary Matricies

The type 1 elementary matrix E_1 is obtained by interchanging two rows. The type 2 elementary matrix E_2 is obtained by multiplying a row by a nonzero scalar k. The type 3 elementary matrix E_3 is obtained by adding a multiple of one row to another row.

- $\det(I) = 1$
- $\det(E_1) = -1$
- $\det(E_2) = k$
- $\det(E_3) = 1$

Determinants of Triangular Matrices

Definition

- $A_{n\times n}$ is called a lower triangular matrix if all entries above the main diagonal are zeros.
- A is called an upper triangular matrix if all entries below the main diagonal are zeros.
- A is a triangular matrix if it is either an upper or lower triangular matrix.
- A is a diagonal matrix if all entries above and below the main diagonal are zeros.
- If A is diagnonal, then it is also triangular.

Theorem If $A_{n\times n}$ is triangular, then $\det(A) = a_{11}a_{22}\cdots a_{nn}$.

The Geometric Meaning of the Determinant

1. $\mathbb{R}^2$ Let $A_{n\times n}$ and $T_A(\vec{x}) = A\vec{x}$. In $\mathbb{R}^2$, $\det(A)$ gives the area-scaling factor of T_A This means if S is a region in $\mathbb{R}$, then $Area(T_A(S)) = |\det(A)| \cdot Area(S)$. The area of the parallelogram spanned by vectors $\vec{u}$ and $\vec{v}$ is given by $A = |\det([\vec{u}, \vec{v}])|$

2. $\mathbb{R}^3$ If B is a solid in $\mathbb{R}^3$, $vol(T_A(B)) = |\det(A)|vol(B)$ This means that the volume of the parallelepiped spanned by $\vec{u}, \vec{v}, \vec{w}$ is given by $V = |\det([\vec{u}, \vec{v}, \vec{w}])|$

Cross Product The cross product $\vec{u} \times \vec{v}$ is the vector orthogonal to both $\vec{u}$ and $\vec{v}$ with magnitude $|\vec{u}||\vec{v}|\sin(\theta)$. It is only defined on $\mathbb{R}^3$ and is given by:

$$\vec{u} \times \vec{v} = \det \begin{vmatrix} \hat{i} & \hat{(j)} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \hat{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Eigenvalues and Eigenvectors

Definition Let $A \in \mathbb{R}^{n\times n}$ and $\vec{x} \in \mathbb{R}^n$ be a nonzero vector. If $A\vec{x} = \lambda\vec{x}$ for some scalar λ, then

- λ is called an eigenvalue of A
- $\vec{x}$ is called an eigenvector of A corresponding to λ

Characteristic Polynomial The characteristic polynomial of A is defined as $P_A(\lambda) = \det(A - \lambda I)$, where I is the $n \times n$ identity matrix. The roots of $P_A(\lambda)$ are the eigenvalues of A. The eigenvectors corresponding to an eigenvalue λ are the nonzero solutions to the homogeneous system $(A - \lambda I)\vec{x} = \vec{0}$. The set of all eigenvectors corresponding to λ together with the zero vector is called the eigenspace of A corresponding to λ.

Theorem Let $A \in \mathbb{R}^{n\times n}$

1. The eigenvalues λ of A are the roots of the characteristic olynomial $P_A(\lambda) = \det(A - \lambda I)$
2. The λ-eigenvectors $\vec{x}$ are the non zero solutions to the homogeneous system $(A - \lambda I)\vec{x}$

Example Find the eigenvector related to each eigenvalue:

$$A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}, \lambda_1 = 4, \lambda_2 = -2$$

$$(A - \lambda_1 I) = \begin{bmatrix} 3-4 & 5 \\ 1 & -1-4 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$(A - \lambda_2 I) = \begin{bmatrix} 5 & 5 \\ 1 & 1 \end{bmatrix} \vec{x} = \vec{0}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \vec{x} \implies \vec{x} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ is a basic eigenvector of A corresponding to } \lambda_2 = -2$$

Transposition

Theorem $A \in \mathbb{R}^{n\times n}$ and A^T have the same characteristic polynomial. Therefore they have the same eigenvalues. $A\vec{x} = \lambda\vec{x} \wedge A^T\vec{x} = \lambda\vec{x}$

Example Find the a. characteristic polynomial b. eigenvalues c. eigenvectors for $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$

$$P_A(\lambda) = \det(A - \lambda I) = \det \begin{vmatrix} 2-\lambda & 0 & 0 \\ 1 & 2-\lambda & -1 \\ 1 & 3 & -2-\lambda \end{vmatrix} = (2-\lambda)\det \begin{vmatrix} 2-\lambda & -1 \\ 3 & -2-\lambda \end{vmatrix} = (2-\lambda)((2-\lambda)(-2-\lambda)+3)$$

$$(A-2I) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & 3 & -4 \end{bmatrix} \xmapsto{RREF} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{x} = t\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ is a}$$

$$(A-I) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 3 & -3 \end{bmatrix} \xmapsto{RREF} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{x} = t\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ is a}$$

$$(A+I) = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & -1 \\ 1 & 3 & -1 \end{bmatrix} \xmapsto{RREF} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{x} = t\begin{bmatrix} 0 \\ \frac{1}{3} \\ 1 \end{bmatrix} \text{ is a}$$

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3.3 Diagonalization

Definition A matrix $A \in \mathbb{R}^{n\times n}$ is diagonalizable if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}$$

Diagonal and triangular matricies have their eigenvalues on the main diagonal.

Theorem An $n \times n$ matrix A is said to be diagonalizable if and only if $P^{-1}AP = D$ whe P is some invertible matrix and D is a diagonal matrix. In this case, the diagonal entries of D are the eigenvalues of A and the columns of P are the corresponding eigenvectors of A.

Properties

- $C = \text{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$ is diagonal $\implies$ C is diagonalizable
- $CD = \text{diag}(c_1d_1, c_2d_2, \cdots, c_nd_n) = CD$ is diagonal $\implies$ CD is diagonalizable

$$A = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Theorem An $n \times n$ matrix A has n eigenvalues

- Not necessarily distinct
- Not necessarily real
- $\det(A) = \lambda_1\lambda_2\cdots\lambda_n$
- $\lambda = 0 \implies \det(A) = 0 \implies A$ is not invertible

Similarity

Definition Two $n \times n$ matrices A and B are similar ($A \sim B$) if there exists an invertible matrix P such that $B = P^{-1}AP$. Similar matrices have the same characteristic polynomial and therefore the same eigenvalues.

How do we find P and D? We want to find P such that $P^{-1}AP = D$ where D is diagonal.

$$P^{-1}AP = D \implies AP = PD$$

$$P = [\vec{v_1}\vec{v_2}\cdots\vec{v_n}] \implies A[\vec{v_1}\vec{v_2}\cdots\vec{v_n}] = [\vec{v_1}\vec{v_2}\cdots\vec{v_n}] \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

$$\implies [A\vec{v_1}A\vec{v_2}\cdots A\vec{v_n}] = [\lambda_1\vec{v_1}\lambda_2\vec{v_2}\cdots\lambda_n\vec{v_n}]$$

$$\implies A\vec{v_i} = \lambda_i\vec{v_i} \text{ for } i = 1, 2, \cdots, n$$

$$\therefore \quad \vec{v_1}, \vec{v_2}, \cdots, \vec{v_n} \text{ are eigenvectors of } A \text{ corresponding to } \lambda_1, \lambda_2, \cdots, \lambda_n$$

Theorem

1. A is diagonalizable $\iff$ it has eigenvectors $\vec{x_1}, \vec{x_2}, \cdots, \vec{x_n}$ and $P = [\vec{x_1}\vec{x_2}\cdots\vec{x_n}]$ is invertible
2. When this is the case, $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$ where λ_i is the eigenvalue corresponding to $\vec{x_i}$ for $i = 1, 2, \cdots, n$
3. If A has n *distinct* eigenvalues, then A is diagonalizable. (Solve $(A - \lambda_i I)\vec{x} = \vec{0}$ for each eigenvalue λ_i to get n linearly independent eigenvectors.)

Example Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$ and $\lambda_1 = 2, lambda_2 = 1, \lambda_3 = -1$. Find a basic eigenvector corresponding to each eigenvalue and determine if A is diagonalizable.

$$(A - 2I) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & 3 & -4 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ is a basic eigenvector corresponding to } \lambda_1 = 2$$

$$(A - I) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 3 & -1 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ is a basic eigenvector corresponding to } \lambda_2 = 1$$

$$(A + I) = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & -1 \\ 1 & 3 & -1 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \text{ is a basic eigenvector corresponding to } \lambda_3 = -1$$

$$\implies P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 4 \end{bmatrix} \text{ and} D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Example Diagonalize $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

1. Charateristic polynomial: $\det(A - \lambda I) = \det \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda^3 + 3\lambda + 2 = -(\lambda - 2)(\lambda + 1)^2$
2. Eigenvalues: $\lambda_1 = 2, \lambda_2 = -1$

3. Eigenvectors:

$$(A-2I) = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$(A+I) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \vec{x} = \vec{0} \implies x_1 + x_2 + x_3 = 0 \implies \vec{x} = t_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\therefore \quad D = \operatorname{diag}(-1,-1,2) \text{ and } P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \text{ is the diagonalized form of } A$$

Multiplicity of Eigenvalues

Definition An eighenvalue λ of $A_{n\times n}$ has multiplicity k if it occurs k times as a root of the characteristic polynomial of A. For example: $P_A(\lambda) = (\lambda-1)^2(\lambda+2)^3$ has eigenvalues $\lambda_1 = 1$ with multiplicity 2 and $\lambda_2 = -2$ with multiplicity 3.

Theorem An $n \times n$ matrix A is diagonalizable if and only if every eigenvalue λ_i of multiplicity m yeilds *exactly* m basic eigenvectors. In particular, if $A_{n\times n}$ has n distinct eigenvalues, then A is diagonalizable.

Diagonalization Algorithm

1. Find the eigenvalues of A by solving $\det(A - \lambda I) = 0$
2. Compute the basic eigenvectors corresponding to each eigenvalue λ_i by solving $(A - \lambda_i I)\vec{x} = \vec{0}$
3. A is diagonalizable $\iff$ there are n basic eigenvectors total. Exactly the multiplicity amount corresponding to each eigenvalue.
4. In this case, the matrix P is formed with these n *basic* eigenvectors as columns and $D = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$ where λ_i is the eigenvalue corresponding to the ith column of P for $i = 1, 2, \cdots, n$.

Example Show that $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is not diagonalizable.

$$\lambda = 1 \text{ with multiplicity } 2 \implies (A - I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\therefore$ There is only 1 basic eigenvector corresponding to the eigenvalue 1 with multiplicity 2

$\implies$ A is not diagonalizable

Example Find the diagonalization of $A = \begin{bmatrix} 1 & 4 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

1. Find eigenvalues:

$$\det(A-\lambda I) = \det \begin{vmatrix} 1-\lambda & 4 & 0 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = (3-\lambda)\det \begin{vmatrix} 1-\lambda & 4 \\ 1 & 1-\lambda \end{vmatrix} = (3-\lambda)((1-\lambda)^2 - 4)$$

$$= (3-\lambda)(\lambda^2 - 2\lambda - 3) = (3-\lambda)(\lambda-3)(\lambda+1) = -(3-\lambda)^2(\lambda+1)$$

2. Find basic eigenvectors:

$$(A-3I) = \begin{bmatrix} -2 & 4 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$(A+I) = \begin{bmatrix} 0 & 4 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore \quad D = \text{diag}(3,3,-1) \text{ and } P = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is the diagonalized form of } A$$

3. Find A^3 using the diagonalization of A:

$$A^3 = PD^3P^{-1} = P\begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & -1 \end{bmatrix} P^{-1} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 54 & -54 & 0 \\ 27 & -27 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Basic Intro to Jordan Cononical Form

Definition A matrix A is in Jordan cononical form if it is a block diagonal matrix where each block is a Jordan block. A Jordan block is a square matrix of the form

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{bmatrix}_{k\times k}$$

Theorem $J_k(\lambda)$ is not diagonalizable for $k > 1$ because it has only one eigenvalue λ with multiplicity k but only one basic eigenvector.

Theorem $\forall A \sim J, J = \text{diag}(J_{k_1}(\lambda_1), J_{k_2}(\lambda_2), \cdots, J_{k_m}(\lambda_m))$ where $J_{k_i}(\lambda_i)$ is a Jordan block corresponding to the eigenvalue λ_i for $i = 1, 2, \cdots, m$.

Example $J = \text{diag}(J_1(5), J_2(7), J_1(7), J_3(8))$, find the dimensions of J

$$J \in \mathbb{R}^{n\times n} \implies n = 1 + 2 + 1 + 3 = 7$$

Theorem $P^{-1}AP = J = \text{diag}(J_{k_1}(\lambda_1), J_{k_2}(\lambda_2), \cdots, J_{k_m}(\lambda_m))$ where $J_{k_i}(\lambda_i)$ is called a block diagonal. A is diagonalizable $\iff$ every Jordan block has size 1. Otherwise, if there is a Jordan block of $P^{-1}AP$ that is larger than 1×1, then A is not diagonalizable.

Example Show $J_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$ is not diagonalizable.

1. $J_3(\lambda)$ has only one eigenvalue λ with multiplicity 3
2. $A - \lambda I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \vec{x} = \vec{0} \implies \vec{x} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
3. There is only one basic eigenvector corresponding to the eigenvalue λ with multiplicity 3 $\implies$ $J_3(\lambda)$ is not diagonalizable

4. Skipped

5. Section 5

5.1 Subspaces and Spanning Sets

Subspaces

Definition A set $U \in \mathbb{R}^n$ is a *subspace* of $\mathbb{R}^n$ if and only if:

1. $\vec{0} \in U$, or U cannot be empty
2. $\vec{x}, \vec{y} \in U \implies \vec{x} + \vec{y} \in U$, or U is closed under vector addition
3. $\vec{x} \in U \implies \forall k \in \mathbb{R}, k\vec{x} \in U$, or U is closed under scalar multiplication

Properties

1. $U = \{\vec{0}\} \subseteq \mathbb{R}^n$ is a subspace of $\mathbb{R}^n$
2. $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$
3. Any subspace of $\mathbb{R}^n$ other than $\{\vec{0}\}$ and $\mathbb{R}^n$ is a *proper subspace* of $\mathbb{R}^n$
4. Any line through the origin in $\mathbb{R}^2$ is a proper subspace of $\mathbb{R}^2$.
    - $\ell = \begin{bmatrix} x \\ mx \end{bmatrix}$ for some $x \in \mathbb{R}$
    - $\vec{0} \in \ell$
    - $\begin{bmatrix} x_1 \\ mx_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ mx_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ m(x_1 + x_2) \end{bmatrix} \in \ell$
    - $k \begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} kx \\ kmx \end{bmatrix} \in \ell$
    - $\therefore$, ℓ is a subspace of $\mathbb{R}^2$
5. Lines are the *only* proper subspaces of $\mathbb{R}^2$
6. Lines and planes through the origin are the *only* proper subspaces of $\mathbb{R}^3$

Is $M = \{\vec{n} \cdot \vec{m} = 0 | \vec{p} \in \mathbb{R}^3\}$, is M a subspace of $\mathbb{R}^3$?

1. $\vec{n} \cdot \vec{0} = 0 \implies \vec{0} \in M$
2. $\vec{u}, \vec{v} \in M \implies \vec{n} \cdot (\vec{u} + \vec{v}) = \vec{n} \cdot \vec{u} + \vec{n} \cdot \vec{v} = 0 + 0 = 0 \implies \vec{u} + \vec{v} \in M$
3. $\vec{v} \in M \implies \vec{n} \cdot (k\vec{v}) = k(\vec{n} \cdot \vec{v}) = k \cdot 0 = 0 \implies k\vec{v} \in M$

Null Space and Image Space

Definition Let A be an $m \times n$ matrix. Then

- $\text{Null}(A) = \{\vec{x} \in \mathbb{R}^n | A\vec{x} = \vec{0}\}$ is the *null space* of A. This is the set of all solutions to the homogeneous system $A\vec{x} = \vec{0}$, and is a subspace of $\mathbb{R}^n$.
- $\text{Image}(A) = \{A\vec{x} | \vec{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$ is the *image space* of A. This is the set of all vectors that can be expressed as a linear combination of the columns of A, and is a subspace of $\mathbb{R}^m$. Also known as the *column space* of A.
- The image set can also be described as all vectors $\vec{b} \in \mathbb{R}^m$ such that $A\vec{x} = \vec{b}$ is consistent.

Eigenspace The *eigenspace* of a matrix A corresponding to an eigenvalue λ is the null space of $A - \lambda I$. It is the set of all eigenvectors corresponding to λ, along with the zero vector. It is a subspace of $\mathbb{R}^n$.

Example Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$. This implies $\lambda_1 = 1$ and $\lambda_2 = 2$.

1. Find $E_{\lambda_1}(A)$

$$\begin{aligned} E_{\lambda_1}(A) &= \text{Null}(A - \lambda_1 I) \\ &= \text{Null}\left(\begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix}\right) = \{[t, 0] \in \mathbb{R}^2 | t \in \mathbb{R}\} \end{aligned}$$

5.2 Dimension

Defintion If U is a subspace of $\mathbb{R}^n$, then a set $B = \{\vec{x}_1, \cdots, \vec{x}_k\} \subseteq \mathbb{R}^n$ is called a **basis** for U if

1. B is linearly independent, and
2. B spans U.

Theorem If $\{\vec{x}_1, \cdots, \vec{x}_m\}$ and $\{\vec{y}_1, \cdots, \vec{y}_p\}$ are both bases for a subspace U of $\mathbb{R}^n$, then $m = p$.

Definition If $\{\vec{x}_1, \cdots, \vec{x}_k\}$ is a basis for a subspace U of $\mathbb{R}^n$, then the **dimension** of U, denoted $\dim(U)$, is defined to be k, the number of vectors in the basis.

Example $\dim \mathbb{R}^n = n \implies$ a basis of $\mathbb{R}^n$ has n vectors, for example $\{\vec{e}_1, \cdots, \vec{e}_n\}$, the standard unit vectors in $\mathbb{R}^n$.

Properties

- You need *at least* n vectors to span $\mathbb{R}^n$.
- You can have *at most* n linearly independent vectors in $\mathbb{R}^n$.
- Any set of n linearly independent vectors in $\mathbb{R}^n$ forms a basis for $\mathbb{R}^n$.
- Any set of n vectors that spans $\mathbb{R}^n$ forms a basis for $\mathbb{R}^n$.

Example Let $U = \{\begin{bmatrix} r \\ s \\ r \end{bmatrix} \mid r, s \in \mathbb{R}\}$. Show that U is a subspace of $\mathbb{R}^3$ and find a basis and the dimension of U.

$$\begin{bmatrix} r \\ s \\ r \end{bmatrix} = r\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \implies U = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}$$

$$\implies U \text{ has a basis } \left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}$$

$$\implies \dim(U) = 2$$

Example Let $V = \left\{\begin{bmatrix} a+b \\ 2a-c \\ c+3a \\ a-b \end{bmatrix} \mid a, b, c \in \mathbb{R}\right\}$. Is V a subspace of $\mathbb{R}^4$? If so, find a basis and the dimension of V.

$$\implies a\begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} + b\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} + c\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} \implies B = \left\{\begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}\right\}$$ spans V and all elements of B are linearly in

Theorem Any subspace $U \neq \{\vec{0}\}$ has a basis and the dimension of U is constant.

Theorem Let $B = \{\vec{x}_1, \cdots, \vec{x}_k\} \subseteq \mathbb{R}^n$ be a basis for a subspace U. If $A_{n\times n}$ is invertible, then $B_A = \{A\vec{x}_1, \cdots, A\vec{x}_k\}$ is also a basis for $U_A = \{A\vec{u} \mid \vec{u} \in U\}$.

Theorems/Facts If $U \neq \{\vec{0}\}$ is a subspace of $\mathbb{R}^n$, then:

1. U has a basis and $\dim(U) \leq n$.
2. Any independent set B in U can be extended to a basis for U by adding vectors that are independent of elements of B. We can use elements from any fixed basis for U to extend B to a basis.
3. Any spanning set B for U can be cut down to a basis of U by removing vectors that are linear combinations of the others in B until the remaining set is linearly independent and still spans U.
4. Check for linear independence by finding determinant and checking that it is nonzero (for a square matrix formed from the vectors as columns).

Example Find a basis for $\mathbb{R}^4$ containing $B = \{\vec{u}, \vec{v}\}$ where $\vec{u} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ We can extend B by adding $\vec{e}_1$ and $\vec{e}_2$ to get a basis for $\mathbb{R}^4$:

$$B' = \{\vec{u}, \vec{v}, \vec{e}_1, \vec{e}_2\} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Example Let $A = \begin{bmatrix} 1 & 3 & -1 & 7 \\ 2 & 6 & -2 & 14 \end{bmatrix}$.

$$\mathrm{IM}(A) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 7 \\ 14 \end{bmatrix} \right\}$$

$$\mathrm{RREF}(A) = \begin{bmatrix} 1 & 3 & -1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies \mathrm{IM}(A) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\} \text{ and } \dim(\mathrm{IM}(A)) = 1$$

Note When finding the basis for the image of a vector, reduce it to RREF and take the pivot columns of the original matrix as the vectors that form a basis for the image.

5.3 Vector Projections

Definition The *vector projection* of a vector $\vec{v}$ onto a nonzero vector $\vec{u}$ is the vector

$$\text{proj}_{\vec{u}}\vec{v} = \left(\frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\right)\vec{u}$$

This vector is parallel to $\vec{u}$ and represents the component of $\vec{v}$ in the direction of $\vec{u}$.

Geometric Interpretation Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$. The *expansion* of $\vec{x} = \sum_{i=1}^n x_i\vec{e}_i$ is the representation of $\vec{x}$ as a linear combination of the standard basis vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$. The coefficients x_i are the components of $\vec{x}$ in the direction of the corresponding basis vectors.

Say we have a different basis $B = \{\vec{f}_1, \vec{f}_2, \ldots, \vec{f}_n\}$ for $\mathbb{R}^n$. We can also express $\vec{x}$ as a linear combination of the vectors in B: $\vec{x} = t_1\vec{f}_1 + t_2\vec{f}_2 + \cdots + t_n\vec{f}_n$. We say that all t_i are the coordinates of $\vec{x}$ relative to the basis B.

This is used in computer graphics, where we often project vectors onto different bases to perform transformations, such as rotations, scaling, and camera position.

Expansion Theorem Let $B = \{\vec{f}_1, \vec{f}_2, \ldots, \vec{f}_n\}$ be a basis for $\mathbb{R}^n$. Then for any vector $\vec{x} \in \mathbb{R}^n$, there exist unique scalars $t_1, t_2, \ldots, t_n$ such that

$$\vec{x} = t_1\vec{f}_1 + t_2\vec{f}_2 + \cdots + t_n\vec{f}_n$$

The scalars t_i are the coordinates of $\vec{x}$ relative to the basis B.

We can find the coordinates t_i by taking the vector projection of $\vec{x}$ onto each basis vector $\vec{f}_i$: $\vec{X} = \sum_{i=1}^n \text{proj}_{\vec{f}_i}\vec{x} = \sum_{i=1}^n \left(\frac{\vec{x}\cdot\vec{f}_i}{\vec{f}_i\cdot\vec{f}_i}\right)\vec{f}_i$. This is called the *Fourier expansion* of $\vec{x}$ relative to the basis B.

Example Let $B = \{\vec{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \vec{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \vec{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \vec{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}\}$ be an orthogonal basis for $\mathbb{R}^4$ and $\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \in \mathbb{R}^4$. Find the coordinates of $\vec{x}$ relative to the basis B, aka $\vec{x}_B$.

$$t_1 = \frac{\vec{x}\cdot\vec{f}_1}{||\vec{f}_1||^2} = \frac{1\cdot 1 + 2\cdot 1 + 3\cdot 1 + 4\cdot(-1)}{1^2 + 1^2 + 1^2 + (-1)^2} = \frac{2}{4} = \frac{1}{2}$$

$$t_2 = \frac{\vec{x}\cdot\vec{f}_2}{||\vec{f}_2||^2} = \frac{1\cdot 1 + 2\cdot 0 + 3\cdot 1 + 4\cdot 2}{1^2 + 0^2 + 1^2 + 2^2} = \frac{12}{6} = 2$$

$$t_3 = \frac{\vec{x}\cdot\vec{f}_3}{||\vec{f}_3||^2} = \frac{1\cdot(-1) + 2\cdot 0 + 3\cdot 1 + 4\cdot 0}{(-1)^2 + 0^2 + 1^2 + 0^2} = \frac{2}{2} = 1$$

$$t_4 = \frac{\vec{x}\cdot\vec{f}_4}{||\vec{f}_4||^2} = \frac{1\cdot(-1) + 2\cdot 3 + 3\cdot(-1) + 4\cdot 1}{(-1)^2 + 3^2 + (-1)^2 + 1^2} = \frac{6}{12} = \frac{1}{2} \therefore \qquad \vec{x}_B = \begin{bmatrix} \frac{1}{2} \\ 2 \\ 1 \\ \frac{1}{2} \end{bmatrix}$$

Example What is $\vec{y}$ if $\vec{y}_B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 3 \end{bmatrix}$?

$$\begin{aligned} \vec{y} &= 1 \cdot \vec{f_1} + 0 \cdot \vec{f_2} + 0 \cdot \vec{f_3} + 3 \cdot \vec{f_4} \\ &= \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 10 \\ -2 \\ 2 \end{bmatrix} \end{aligned}$$

Theorem If $A \in \mathbb{R}^{m \times n}$ has orthogonal columns, then

$$A^T A = D = \text{diag}(||\vec{c}_1||^2, ||\vec{c}_2||^2, \dots, ||\vec{c}_n||^2) \implies ||\vec{c}_i||^2 = d_i$$

where $\vec{c}_i$ is the i-th column of A. Remember that matricies are not transitive, so $A^T A \neq AA^T$ in general.

Theorem Let $Q \in \mathbb{R}^{n \times n}$. If $Q^T Q = I$, then the columns of Q are orthonormal, meaning the vectors have unit length and are mutually perpendicular. Conversely, if the columns of Q are orthonormal, then $Q^T Q = I$.

Theorem If S is orthogonal and $\vec{0} \notin S$, then S is linearly independent. This is because the columns of S are orthogonal, and therefore cannot be linear combinations of each other.

$$\sum i = 1^n t_i \vec{v}_i = \vec{0} \implies \forall \vec{v}_i \in S, \vec{v}_i \text{ is linearly independent of the others} \implies t_i = 0$$

Rank of a Matrix

Definition The *rank* of a matrix A, denoted $\mathrm{rank}(A)$, is the maximum number of linearly independent rows or columns in A. Previously we have seen that the rank of a matrix is equal to the number of pivots in its row echelon form, which is also equal to the number of nonzero rows in its reduced row echelon form. The rank of a matrix can also be defined as the dimension of the column space or the row space of the matrix.

Definition The *column space* of a matrix A, denoted $\mathrm{Col}(A) = \mathrm{span}\{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\} = \mathrm{Im}(A)$, is the set of all linear combinations of the columns of A. The column space is a subspace of $\mathbb{R}^n$ if A is an $m \times n$ matrix.

The *row space* of a matrix A, denoted $\mathrm{Row}(A)$, is the set of all linear combinations of the rows of A. The row space is a subspace of $\mathbb{R}^m$ if A is an $m \times n$ matrix.